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(54) **SYSTEMS AND METHODS FOR
DISPATCHING ELEVATORS**

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(57) **ABSTRACT**

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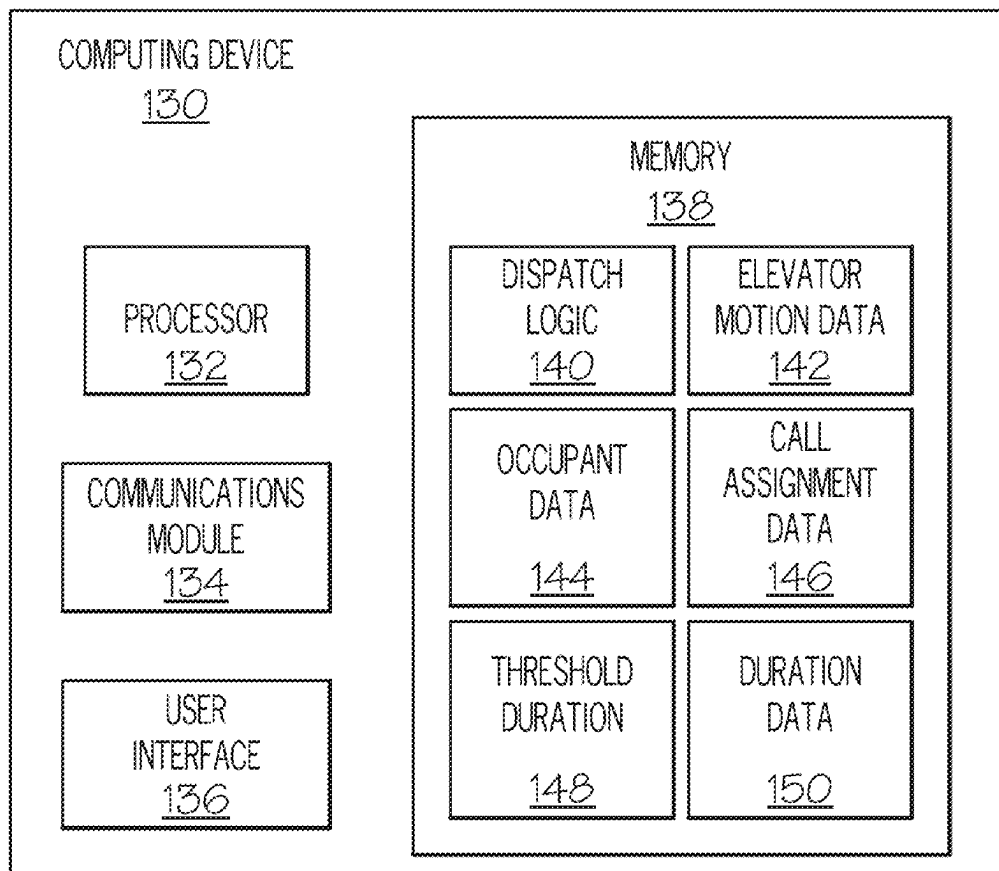
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A method for dispatching an elevator car that includes
determining a first duration for each of a plurality of first
elevator cars to travel from a current location to a first
location, and dispatching at least one of the first elevator cars
to the first location when the first duration of the at least one
first elevator car is less than a threshold duration. The
method includes determining a second duration for an
occupant to travel from the first location to a second loca-
tion, and each of a plurality of second elevator cars to travel
from the current location to the second location, when the
first duration for the first elevator cars exceeds the threshold
duration. The method includes dispatching at least one of the
second elevator cars to the second location when the second
duration of the at least one second elevator car is less than
the threshold duration.



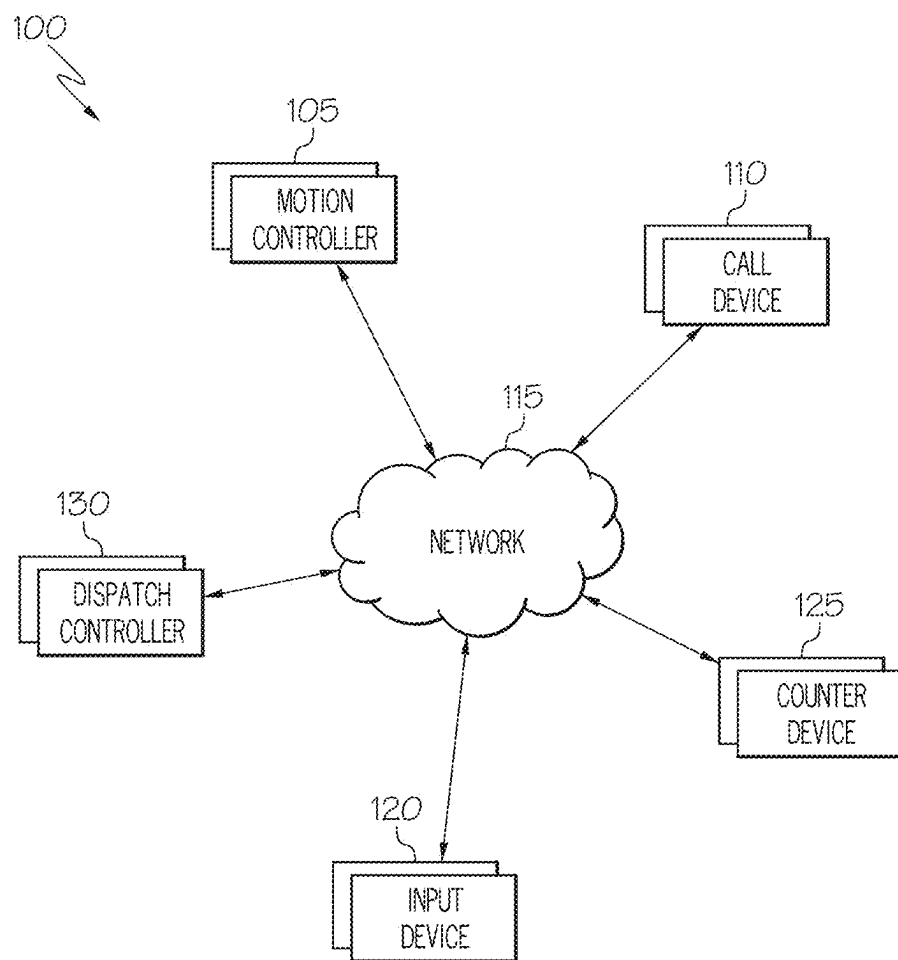


FIG. 1

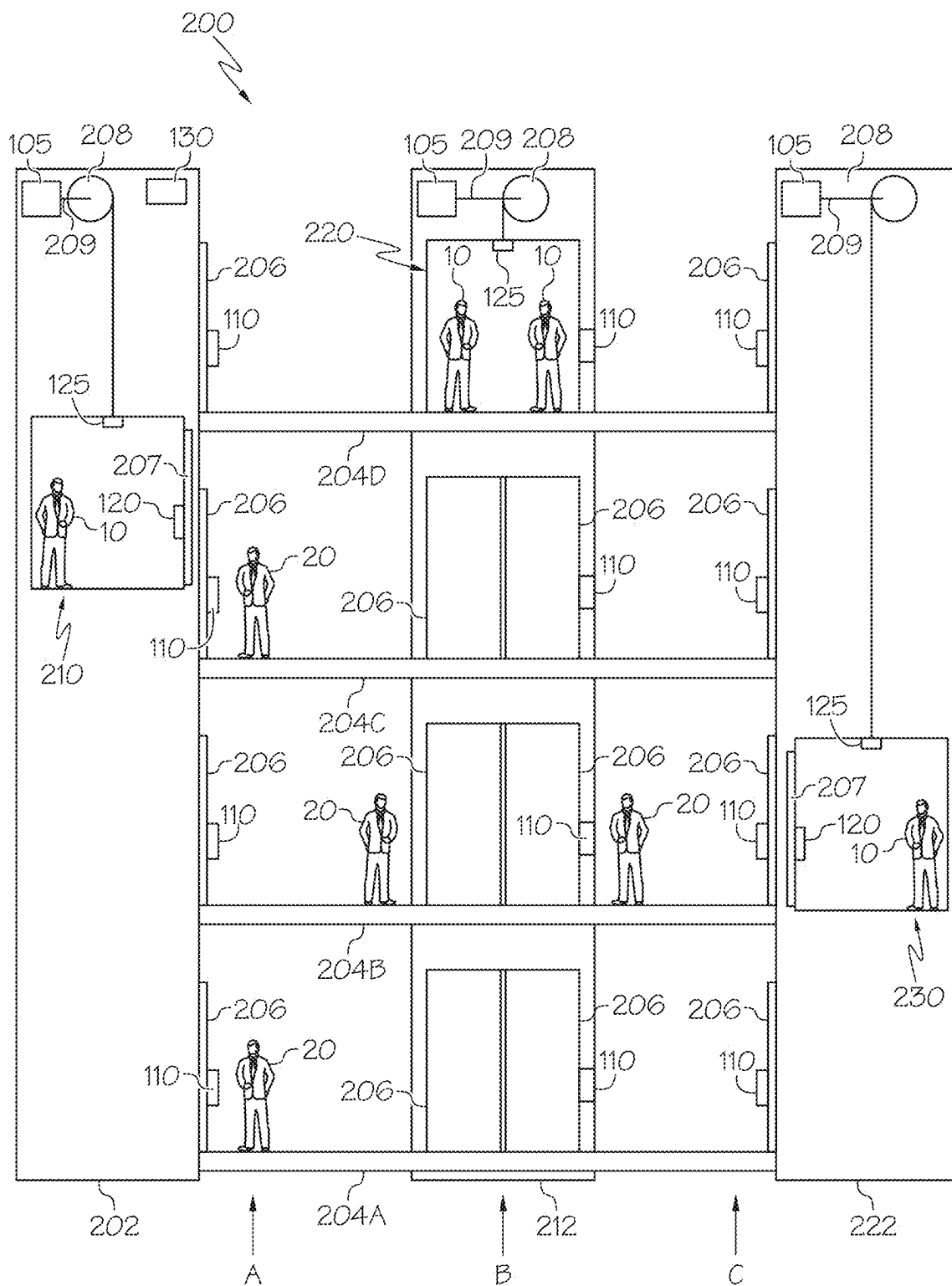


FIG. 2

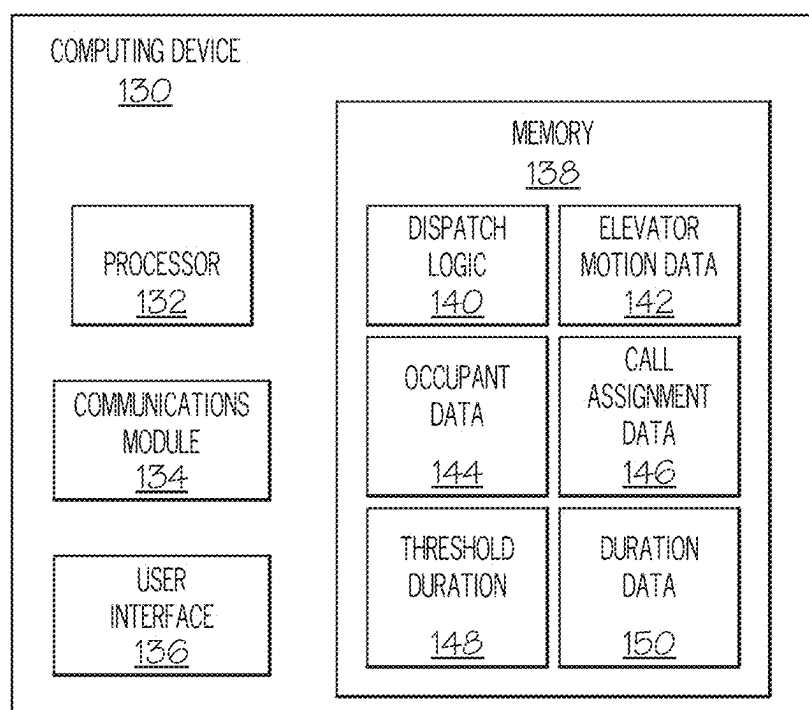
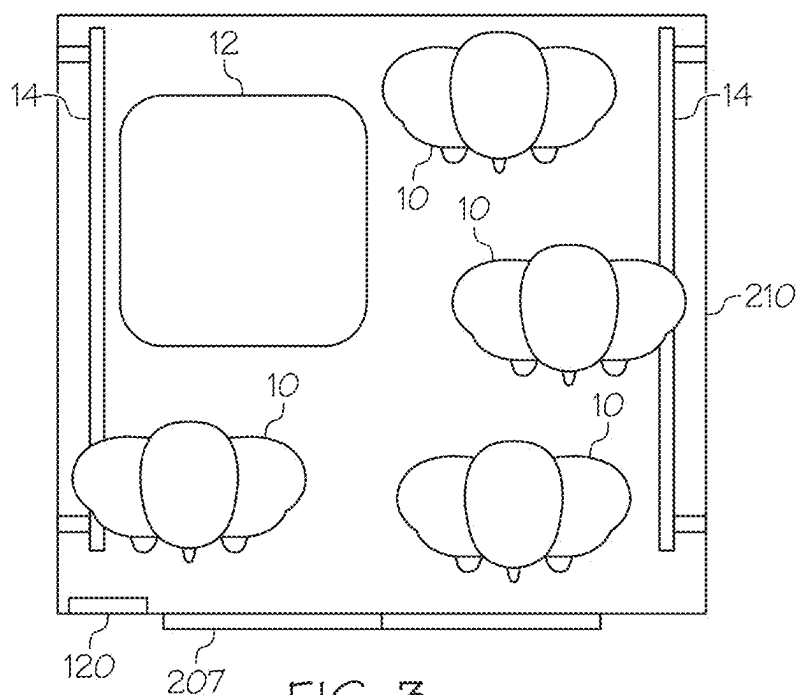
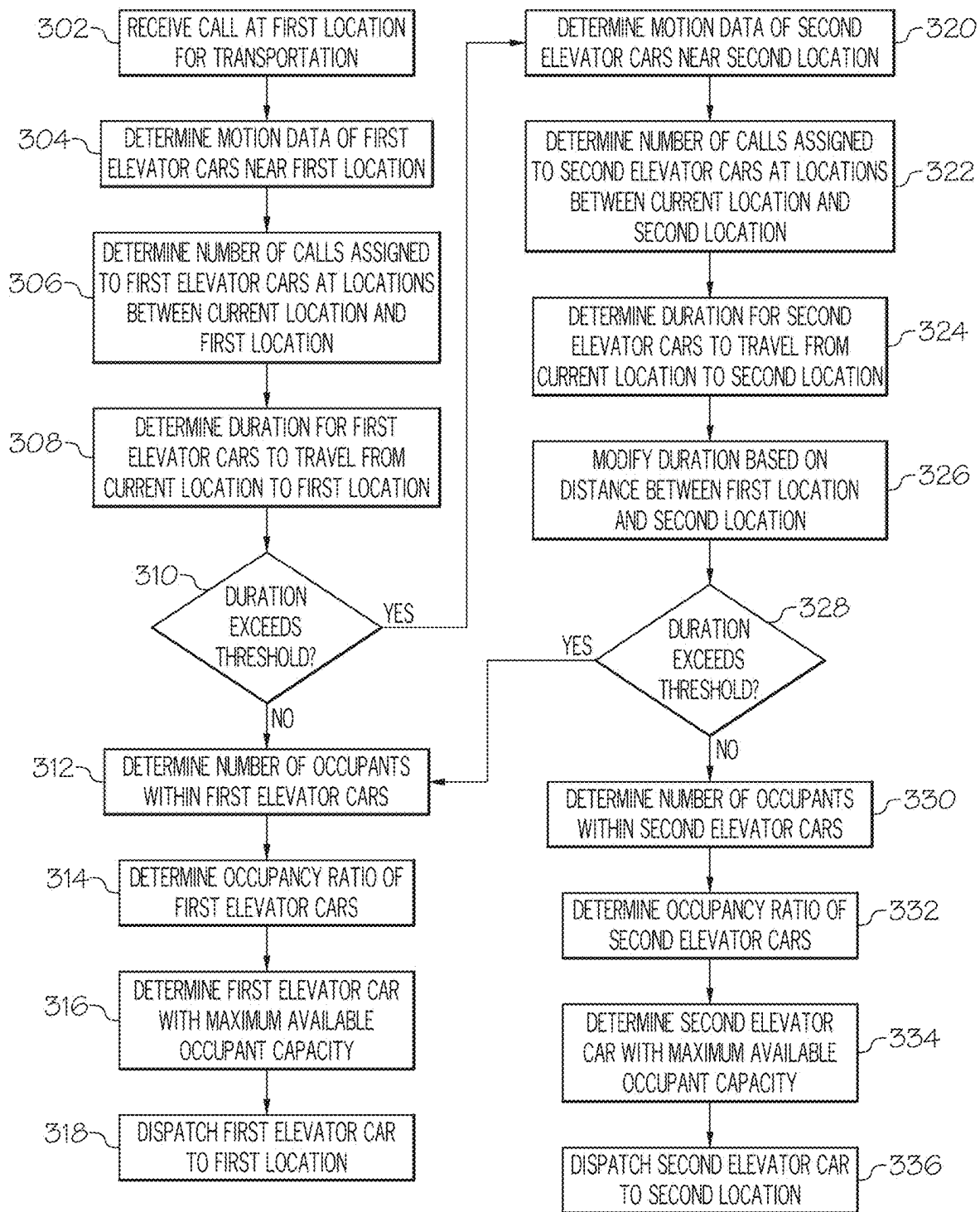


FIG. 4



300
FIG. 5

SYSTEMS AND METHODS FOR DISPATCHING ELEVATORS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 18/936,429, filed Nov. 4, 2024, which is a continuation of U.S. application Ser. No. 17/486,502, filed on Sep. 27, 2021, which claims the benefit of priority from U.S. Provisional Application No. 63/084,452, filed on Sep. 28, 2020, which are incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] Aspects of the present disclosure relate generally to systems and methods for controlling elevator traffic flow, and specifically to examples of elevator control systems that dispatch elevator cars based on a travel duration relative to a group of elevator cars.

DESCRIPTION OF RELATED TECHNOLOGY

[0003] Elevator systems may generally employ a dispatch methodology based on a necessary travel time to answer a call request. In such systems, an estimated travel time of each elevator car may be determined when a call request is received. An elevator car located near the call location, and having the smallest travel time to the call location, may be dispatched to a location of the call request. In some instances, an elevator car located at another location may have the smallest travel time, such that said elevator car may be dispatched to the call request. The prospective passenger initiating the call request may be instructed to travel to the other location for pick up.

[0004] However, assigning elevator cars that are located at distant locations (relative to where the call request originated from) may result in prospective passengers missing the elevator car upon their arrival to the location. This may be due to a duration required for the prospective passenger to travel to the location of the elevator car. As a result, prospective passengers may be required to attempt another call request for a separate elevator car, thereby resulting in decreased traffic flow and greater wait times. Providing a system capable of determining a travel duration between locations when assigning elevator cars may minimize instances of dispatching elevator cars that prospective passengers may not be capable of reaching, thereby increasing traffic flow and decreasing wait times for prospective passengers.

BRIEF DESCRIPTION OF DRAWINGS

[0005] The accompanying drawings, which are incorporated in and constitute a part of this disclosure, illustrate various exemplary embodiments and together with the description, serve to explain the principles of the disclosure.

[0006] Aspects of the disclosure may be implemented in connection with embodiments illustrated in the attached drawings. These drawings show different aspects of the present disclosure and, where appropriate, reference numerals illustrating like structures, components, materials and/or elements in different figures are labeled similarly. It is understood that various combinations of the structures, components, and/or elements, other than those specifically shown, are contemplated and are within the scope of the

present disclosure. There are many aspects and embodiments described herein. Those of ordinary skill in the art will readily recognize that the features of a particular aspect or embodiment may be used in conjunction with the features of any or all of the other aspects or embodiments described in this disclosure.

[0007] FIG. 1 depicts a dispatch system including one or more devices in communication over a network.

[0008] FIG. 2 is a schematic view of a working environment including multiple elevator cars at different locations interacting with the dispatch system shown in FIG. 1.

[0009] FIG. 3 is a top view of an interior of an elevator car from the working environment shown in FIG. 2.

[0010] FIG. 4 is a schematic view of hardware components of a computing device from the dispatch system shown in FIG. 1.

[0011] FIG. 5 is a flow diagram of an exemplary method of dispatching elevator cars with the dispatch system shown in FIG. 1.

SUMMARY

[0012] According to an example, a method for dispatching an elevator car includes determining a first duration for each of a plurality of first elevator cars to travel from a current location to a first location; dispatching at least one of the plurality of first elevator cars to the first location when the first duration of the at least one first elevator car is less than a threshold duration and the first duration of the remaining plurality of first elevator cars; determining a second duration for (i) an occupant to travel from the first location to a second location, and (ii) each of a plurality of second elevator cars to travel from the current location to the second location, when the first duration for each of the plurality of first elevator cars exceeds the threshold duration; and dispatching at least one of the plurality of second elevator cars to the second location when the second duration of the at least one second elevator car is less than the threshold duration and the second duration of the remaining plurality of second elevator cars.

[0013] According to another example, a system for dispatching an elevator car includes at least one motion controller operably coupled to a plurality of elevator cars, the at least one motion controller is configured to determine a current location of the plurality of elevator cars, wherein the plurality of elevator cars includes a first subset and a second subset; and a dispatch controller operably coupled to the at least one motion controller of the plurality of elevator cars, such that the dispatch controller receives data indicative of the current location of the plurality of elevator cars. The dispatch controller is configured to: determine a first duration for each of the first subset of elevator cars to travel from the current location to a first location; dispatch, in response to the first duration of at least one of the first subset of elevator cars not exceeding a threshold duration, the at least one elevator car of the first subset to the first location; determine, in response to the first duration for each of the first subset of elevator cars exceeding the threshold duration, a second duration for (i) an occupant to travel from the first location to a second location, and (ii) each of the second subset of elevator cars to travel from the current location to the second location; and dispatch, in response to the second duration of at least one of the second subset of elevator cars not exceeding the threshold duration, the at least one elevator car of the second subset to the second location.

[0014] According to a further example, a system for controlling traffic flow of a plurality of elevator cars, comprising a processor and a memory storing instructions that, when executed by the processor, causes the processor to perform operations including: determining a first duration for each of a plurality of first elevator cars to travel from a current location to a first location; dispatching at least one of the plurality of first elevator cars to the first location when the first duration of the at least one first elevator car is less than a threshold duration; determining a second duration for (i) an occupant to travel from the first location to a second location, and (ii) each of a plurality of second elevator cars to travel from the current location to the second location, when the first duration for each of the plurality of first elevator cars exceeds the threshold duration; and dispatching at least one of the plurality of second elevator cars to the second location when the second duration of the at least one second elevator car is less than the threshold duration.

DETAILED DESCRIPTION

[0015] The dispatch system of the present disclosure may be in the form of varying embodiments, some of which are depicted by the figures and further described below.

[0016] Both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the features, as claimed. As used herein, the terms “comprises,” “comprising,” or other variations thereof, are intended to cover a non-exclusive inclusion such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent to such a process, method, article, or apparatus. Additionally, the term “exemplary” is used herein in the sense of “example,” rather than “ideal.” It should be noted that all numeric values disclosed or claimed herein (including all disclosed values, limits, and ranges) may have a variation of $\pm 10\%$ (unless a different variation is specified) from the disclosed numeric value. Moreover, in the claims, values, limits, and/or ranges mean the value, limit, and/or range $\pm 10\%$.

[0017] FIG. 1 shows an exemplary dispatch system 100 that may include motion controller 105, call device 110, input device 120, counter device 125, and dispatch controller 130. The one or more devices of dispatch system 100 may communicate with one another across a network 115 and in any arrangement. For example, the devices of dispatch system 100 may be communicatively coupled to one another via a wired connection, a wireless connection, or the like. In some embodiments, network 115 may be a wide area network (“WAN”), a local area network (“LAN”), personal area network (“PAN”), etc. Network 115 may further include the Internet such that information and/or data provided between the devices of dispatch system 100 may occur online (e.g., from a location remote from other devices or networks coupled to the Internet). In other embodiments, network 115 may utilize Bluetooth® technology and/or radio waves frequencies.

[0018] Motion controller 105 may be operably coupled to a transportation unit and configured to detect and transmit motion data of the transportation unit to one or more devices of dispatch system 100, such as, for example, dispatch controller 130. For example, motion controller 105 may measure and record one or more parameters (e.g., motion data) of the transportation unit, including, but not limited to,

a current location, a travel direction, a travel speed, a door location, a status, and more. Motion controller 105 may include a computing device having one or more hardware components (e.g., a processor, a memory, a sensor, a communications module, etc.) for generating, storing, and transmitting the motion data. As described in further detail herein, motion controller 105 may be operably coupled to an elevator car located within a building and dispatch system 100 may include at least one motion controller 105 for each elevator car.

[0019] Still referring to FIG. 1, call device 110 may be positioned outside the transportation unit and configured to receive a user input from one or more prospective occupants for accessing the transportation unit. For example, the user input may be indicative of a call requesting transportation from the transportation unit. Call device 110 may be configured to transmit the call request to one or more devices of dispatch system 100, such as, for example, dispatch controller 130. Call device 110 may include a keypad, a touch-screen display, a microphone, a button, a switch, etc. Call device 110 may be further configured to receive a user input indicative of a current location of the call request (e.g., a first location) and/or a destination location from a plurality of locations.

[0020] As described in further detail herein, call device 110 may be located within a building, and dispatch system 100 may include at least one call device 110 for each floor of the building. Call device 110 may be configured to transmit a message from one or more devices of dispatch system 100 (e.g., dispatch controller 130) identifying an elevator car assigned to arrive at the floor of the building to answer the call request. The message may be communicated by call device 110 via various suitable formats, including, for example, in a written form, an audible form, a graphic form, and more.

[0021] Input device 120 may be positioned inside the transportation unit and configured to receive a user input from one or more occupants of the transportation unit. For example, the user input may be indicative of a command requesting redirection of the transportation unit. Input device 120 may be configured to transmit the command to one or more devices of dispatch system 100, such as, for example, dispatch controller 130. Input device 120 may include a keypad, a touchscreen display, a microphone, a button, a switch, etc. As described in detail herein, input device 120 may be located within an elevator car, and dispatch system 100 may include at least one input device 120 for each elevator car in a building. In other embodiments, input device 120 may be omitted entirely from dispatch system 100.

[0022] Still referring to FIG. 1, counter device 125 may be positioned inside the transportation unit and configured to detect and transmit occupant data of the transportation unit to one or more devices of dispatch system 100, such as, for example, dispatch controller 130. For example, counter device 125 may measure and record a number of objects located within the transportation unit, including, but not limited to, an occupant, a personal belonging, a luggage, a baggage, and more. Counter device 125 may include an optical system facing an interior of the transportation unit, such as, for example, a sensor, a camera, a light beam, an infrared detector, etc. As described in further detail herein, counter device 125 may be coupled to an elevator car that is

located within a building, and dispatch system 100 may include at least one counter device 125 for each elevator car of the building.

[0023] Dispatch controller 130 may be positioned outside the transportation unit and configured to receive data (e.g., motion data, a call request, a redirection command, occupant data, etc.) from one or more devices of dispatch system 100. Dispatch controller 130 may be further configured to determine at least one transportation unit of a plurality of transportation units to dispatch in response to a call request received from a prospective passenger seeking transportation. Dispatch controller 130 may include a computing device (see FIG. 4) operable to perform one or more processes (see FIG. 5) for dispatching at least one transportation unit with the smallest duration to pick up a prospective passenger. As described in further detail herein, dispatch controller 130 may be operably coupled to a plurality of elevator cars located within a building, and dispatch system 100 may include at least one dispatch controller 130 for each building.

[0024] Referring now to FIG. 2, dispatch system 100 may be utilized in a working environment 200, such as a building (e.g., a facility, a factory, a store, a school, a house, an office, and various other structures). In the example, the transportation unit may include one or more elevator cars within the building. It should be appreciated that working environment 200 is merely illustrative such that dispatch system 100 may be utilized in various other suitable environments than those shown and described herein without departing from a scope of this disclosure. In the example, working environment 200 may include a plurality of floors defining a plurality of locations within the building, such as first floor 204A, second floor 204B, third floor 204C, and fourth floor 204D. It should be appreciated that, in other embodiments, the building of working environment 200 may include additional and/or fewer floors.

[0025] Working environment 200 may further include one or more elevator shafts with at least one elevator car positioned within each elevator shaft. In the example, working environment 200 includes a first elevator shaft 202 with at least one first elevator car 210, a second elevator shaft 212 with at least one second elevator car 220, and a third elevator shaft 222 with at least one third elevator car 230. Each elevator shaft 202, 212, 222 may be located at a different location on each of the plurality of floors 204A-204D. Stated differently, first elevator shaft 202 may be located at a first location “A,” second elevator shaft 212 may be located at second location “B” that is different than the first location “A,” and third elevator shaft 222 may be located at a third location “C” that is different than the first location “A” and second location “B,” on each of the plurality of floors 204A-204D. Although not shown, it should be appreciated that working environment 200 may include additional (e.g., a plurality) elevator shafts, elevator cars, and locations at which said elevator shafts and elevator cars are located. Accordingly, it should be appreciated that working environment 200 may include a plurality of first elevator shafts 202 including a plurality of first elevator cars 210; a plurality of second elevator shafts 212 including a plurality of second elevator cars 220; a plurality of third elevator shafts 222 including a plurality of third elevator cars 230; and more.

[0026] Each elevator car 210, 220, 230 may be coupled to a pulley system 208 configured to move elevator cars 210, 220, 230 within elevator shafts 202, 212, 222 and relative to

Floors 204A-204D. It should be understood that pulley system 208 may include various mechanical and/or electrical mechanisms for moving elevator cars 210, 220, 230 within elevator shafts 202, 212, 222, including but not limited to, a motor, a cable, a counterweight, a sheave, etc.

[0027] Still referring to FIG. 2, each elevator car 210, 220 may include at least one motion controller 105 operably coupled to pulley system 208, such as, for example, via a wireless connection and/or a wired connection 209. Motion controller 105 may be configured to measure motion data from elevator cars 210, 220 by detecting a relative movement of pulley system 208. Each elevator car 210, 220 may further include at least one input device 120 positioned within a cabin of elevator car 210, 220 for receiving a user input from one or more occupants 10 located within the cabin.

[0028] Each floor 204A-204D may include one or more call devices 110 and access doors 206 at a location of each elevator shaft 202, 212, 222 on said floor 204A-204D. Access doors 206 may provide accessibility to elevator cars 210, 220, 230 when an elevator door 207 of elevator car 210, 220, 230 is aligned with the respective floor 204A-204D. Call device 110 may be configured to receive a user input from one or more prospective occupants 20 located at one of the plurality of locations on one of floors 204A-204D. For example, call device 110 may be configured to receive a user input indicative of a call requesting transportation via at least one of elevator cars 210, 220, 230. Call device 110 may be configured to transmit the call request to dispatch controller 130, which may include data indicative of a current location within working environment 200 from which the call request originated from (e.g., the first location “A” on first floor 204A). The call request may further include data indicative of a destination location within working environment 200 to which the prospective passenger is seeking transportation to (e.g., fourth floor 204D).

[0029] Still referring to FIG. 2, each elevator car 210, 220, 230 may further include at least one counter device 125 positioned within a cabin. Counter device 125 may be positioned along an inner wall (e.g., a ceiling) of each elevator car 210, 220, 230 and configured to detect a number of occupants 10 within the cabin. In some embodiments, counter device 125 may be operable to distinguish between one or more objects detected within elevator cars 210, 220, 230.

[0030] For example, as seen in FIG. 3, counter device 125 may be configured to detect items present within the cabin and occupying a capacity of elevator cars 210, 220, 230 (e.g., occupants 10, ancillary objects 12, etc.) and items within the cabin that may not occupy a capacity of elevator cars 210, 220, 230 (e.g., rails 14, etc.). Counter device 125 may measure a number of items detected within elevator cars 210, 220, 230 and record such measurements as occupant data. As discussed further herein, counter device 125 may be configured to transmit occupant data (e.g., occupant data 144) for each elevator car 210, 220, 230 to dispatch controller 130 via network 115.

[0031] Referring now to FIG. 4, dispatch controller 130 may include a computing device incorporating a plurality of hardware components that allow dispatch controller 130 to receive data (e.g., motion data, call requests, commands, occupant data, etc.), process information (e.g., occupant capacity), and/or execute one or more processes (see FIG. 5). Illustrative hardware components of dispatch controller

130 may include at least one processor **132**, at least one communications module **134**, a user interface **136**, and at least one memory **138**. In some embodiments, dispatch controller **130** may include a computer, a mobile user device, a remote station, a server, a cloud storage, and the like. In the illustrated embodiment, dispatch controller **130** is shown and described herein as a separate device from the other devices of dispatch system **100**, while in other embodiments, one or more aspects of dispatch controller **130** may be integrated with one or more of the other devices of dispatch system **100**. Stated differently, the illustrative hardware components of dispatch controller **130** shown and described herein may be integral with one or more of motion controller **105**, call device **110**, input device **120**, and/or counter device **125**.

[0032] Processor **132** may include any computing device capable of executing machine-readable instructions, which may be stored on a non-transitory computer-readable medium, such as, for example, memory **138**. By way of example, processor **132** may include a controller, an integrated circuit, a microchip, a computer, and/or any other computer processing unit operable to perform calculations and logic operations required to execute a program. As described in detail herein, processor **132** is configured to perform one or more operations in accordance with the instructions stored on memory **138**, such as, for example, dispatch logic **140**.

[0033] Communications module **134** may facilitate communication between dispatch controller **130** and the one or more other devices of dispatch system **100**, such as, for example, via network **115**. User interface **136** may include one or more input and output devices, including one or more input ports and one or more output ports. User interface **136** may include, for example, a keyboard, a mouse, a touchscreen, etc., as input ports. User interface **136** may further include, for example, a monitor, a display, a printer, etc. as output ports. User interface **136** may be configured to receive a user input indicative of various commands, including, but not limited to, a command defining and/or adjusting the threshold duration **148** stored in memory **138**, and more.

[0034] Still referring to FIG. 4, memory **138** may include various programming algorithms and data that support an operation of dispatch system **100**. Memory **138** may include any type of computer readable medium suitable for storing data and algorithms, such as, for example, random access memory (RAM), read only memory (ROM), a flash memory, a hard drive, and/or any device capable of storing machine-readable instructions. Memory **138** may include one or more data sets, including, but not limited to, motion data **142** received from motion controller **105**, occupant data **144** captured from counter device **125**, call assignment data **146** and duration data **150** for each of the plurality of elevator cars **210**, **220**, **230**, and the like. Memory **138** may further include a threshold duration **148** that may be preprogrammed and/or adjustable by a user of dispatch system **100**, such as, for example, via user interface **136**.

[0035] As described further herein, occupant data **144** may include a real-time number of occupants **10** detected within a cabin of each elevator car **210**, **220**, **230** by counter device **125**. Call assignment data **146** may include a call request received from a prospective occupant **20** at one of the plurality of floors **204A-204D**, for transportation by at least one of the plurality of elevator cars **210**, **220**, **230**. Dispatch controller **130** may be configured to store the occupant data

144 in memory **138** and associate the number of occupants **10** with a corresponding elevator car **210**, **220**, **230**. Dispatch controller **130** may be further configured to store the call assignment data **146** in memory **138** to determine a current number of stops assigned to each elevator car **210**, **220**, **230**. As described further herein, dispatch controller **130** may be configured to determine a minimum travel duration for each of the plurality of elevator cars **210**, **220**, **230** based on at least one or more of the motion data **142**, the occupant data **144**, and the call assignment data **146**.

[0036] Further, memory **138** may include a non-transitory computer readable medium that stores machine-readable instructions thereon, such as, dispatch logic **140**. In one example, dispatch logic **140** may include executable instructions that allow dispatch system **100** to determine which elevator car from the plurality of elevator cars **210**, **220**, **230** to dispatch in response to receiving a call request at the first location “A” for transportation to a destination location. Dispatch logic **140** may facilitate determining an occupant capacity of each elevator car **210**, **220**, **230** based on a number of occupants physically present within each elevator car **210**, **220**, **230**. Dispatch logic **140** may further facilitate the determination of a minimum duration (e.g., duration data **150**) for each of the plurality of elevator cars **210**, **220**, **230** to travel to a location based on one or more of the motion data **142**, occupant data **144**, and/or call assignment data **146**. As described in further detail herein, dispatch system **100** may be configured to determine at least one elevator car **210**, **220**, **230** having the least duration (e.g., duration data **150**) to travel to a first location in response to a call request from a prospective occupant **20**.

[0037] Referring now to FIG. 5, an example method **300** of using dispatch system **100** to determine a travel duration of a plurality of elevator cars and to dispatch an elevator car having the shortest travel duration is depicted. It should be understood that the steps shown and described herein, and the sequence in which they are presented, are merely illustrative such that additional and/or fewer steps may be included in various arrangements without departing from a scope of this disclosure.

[0038] At step **302**, dispatch system **100** may receive a call request at the first location “A” of a plurality of locations within working environment **200**. The call request may be initiated in response to a prospective occupant **20** actuating call device **110** at the first location “A,” such as, for example, on first floor **204A** and adjacent to a plurality of first elevator shafts **202**. Call device **110** may transmit the call request to dispatch controller **130** via network **115**, and the call request may include data indicative of the first location “A” from which the call request originated. The call request may further include data indicative of a destination location (e.g., fourth floor **204D**) within working environment **200** to which the prospective occupant **20** seeks to travel.

[0039] At step **304**, with the call request originating from first floor **204A** and adjacent to the plurality of first elevator shafts **202**, dispatch controller **130** may retrieve motion data **142** from a corresponding motion controller **105** of a plurality of first elevator cars **210** located within the plurality of first elevator shafts **202**. Dispatch controller **130** may be configured to determine various movement parameters of the plurality of first elevator cars **210** from the motion data **142**, such as, for example, a current location of each first elevator car **210** relative to a respective first elevator shaft

202, a current travel direction of each first elevator car **210**, and a current travel speed of each first elevator car **210**.

[0040] It should be understood that, in response to determining one or more first elevator cars **210** are not traveling toward the first location “A,” dispatch controller **130** may be configured to disregard the particular first elevator car **210** from further consideration. Stated differently, dispatch controller **130** may determine that any elevator car traveling in a different direction than toward the first location “A” (relative to the current location of the elevator car) may not be an optimal elevator car to answer the call request. In the example, working environment **200** includes first elevator car **210** positioned between fourth floor **204D** and third floor **204C**, and moving toward first floor **204A**.

[0041] In some embodiments, dispatch controller **130** may be further configured to determine whether a current location of the plurality of first elevator cars **210** is located prior to the first location “A,” or whether first elevator cars **210** have moved beyond the first location “A.” Stated differently, dispatch controller **130** may determine that any elevator car that is currently positioned beyond the first location “A” may not be an optimal elevator car to answer the call request. In response to determining one or more of the plurality of first elevator cars **210** are not located before the first location “A,” dispatch controller **130** may be configured to disregard said first elevator cars **210** from further consideration. In the example, as seen in FIG. 2, first elevator car **210** is positioned between fourth floor **204D** and third floor **204C**, such that dispatch controller **130** may determine that first elevator car **210** is currently located before first floor **204A** (e.g., the first location “A”).

[0042] Still referring to FIG. 5, at step **306**, dispatch controller **130** may be configured to determine a number of calls assigned to each of the plurality of first elevator cars **210** (e.g., by dispatch controller **130**) and that have a (pick up) location positioned between a current location of each first elevator car **210** and the first location “A.” Stated differently, dispatch controller **130** may determine how many, if any, intermediate stops each first elevator car **210** is expected to have between its current location and the first location “A” (e.g., first floor **204A**). It should be understood that the number of calls previously assigned to each first elevator car **210** is relative to when the call request (step **302**) is received by dispatch controller **130**.

[0043] It should further be appreciated that any calls previously assigned to first elevator cars **210**, and which do not include a location positioned between the current location and the first location “A,” do not form an intermediate stop. Accordingly, dispatch controller **130** may be configured to disregard any prior calls assigned to first elevator cars **210**, and which have a (pick up) location after the first location “A,” when determining the number of calls at step **306**. In the example, as seen in FIG. 2, first elevator car **210** may include a previously-assigned call at third floor **204C** such that dispatch controller **130** may determine that first elevator car **210** includes one stop between the current location of first elevator car **210** and first floor **204A** (e.g., the first location “A”).

[0044] Still referring to FIG. 5, at step **308**, dispatch controller **130** may be configured to analyze the motion data **142** (step **304**) and the call assignment data **146** (step **306**) collected for the plurality of first elevator cars **210** to determine a minimum duration (e.g., a first duration) for each first elevator car **210** to travel from a respective current

location to the first location “A.” For example, dispatch controller **130** may analyze a travel distance between a current location of each first elevator car **210** and the first location “A” when determining the minimum duration. Dispatch controller **130** may further analyze a travel speed of each first elevator car **210** when determining the minimum duration.

[0045] By further example, dispatch controller **130** may determine that the minimum duration required for each first elevator car **210** to travel to the first location “A” increases based on the number of existing calls assigned to the first elevator car **210**. Stated differently, dispatch controller **130** may assess the number of stops assigned to each first elevator car **210**, and positioned between a current location and the first location “A,” when determining the minimum duration for each first elevator car **210**. For example, dispatch controller **130** may compute a predefined increment (e.g., duration data **150**) to the minimum duration for each stop assigned to a particular first elevator car **210**. The predefined increment may be programmed in memory **138**, and may include various suitable values ranging from about 1 second to about 120 seconds, and particularly about 60 seconds. In some embodiments, a value of the predefined increment may be selectively modified by a user of dispatch system **100**, such as via user interface **136**.

[0046] At step **310**, dispatch controller **130** may be configured to compare the minimum duration of the plurality of first elevator cars **210** to the threshold duration **148**. As described in detail above, the threshold duration **148** may define a maximum allotted duration for an elevator car to travel from a respective current location to the first location “A.” In other words, elevator cars determined to have a minimum duration that exceeds the threshold duration **148** may not be an optimal elevator car to answer the call request. In response to determining two or more of the plurality of first elevator cars **210** have a minimum duration that is less than the threshold duration **148** at step **310**, dispatch controller **130** may be configured to determine a number of occupants **10** located within each of the two or more first elevator cars **210** at step **312**.

[0047] For example, dispatch controller **130** may be configured to determine the number of occupants **10** within each first elevator car **210** (having a minimum duration less than the threshold duration **148**) by retrieving occupant data **144** from a respective counter device **125** of each first elevator car **210**. In some embodiments, counter device **125** may be configured to detect a total number of occupants **10** and/or objects **12** located within first elevator cars **210** (see FIG. 3). Thus, dispatch controller **130** may consider one or more objects **12** detected by counter device **125** when determining the number of occupants **10** at step **312**. Each counter device **125** may transmit a signal to dispatch controller **130** via network **115** indicative of the occupant data **144** for the respective first elevator car **210**.

[0048] Still referring to FIG. 5, at step **314**, dispatch controller **130** may be configured to determine an occupancy ratio of each of the plurality of first elevator cars **210** based on at least the number of occupants **10** within the elevator car (step **312**) and a maximum occupant capacity of each elevator car. In some embodiments, a maximum occupant capacity of the plurality of first elevator cars **210** may be communicated to dispatch controller **130** from counter device **125** via network **115**. In other embodiments, dispatch controller **130** may store the maximum occupant capacity in

memory **138** for each of the plurality of first elevator cars **210**. It should be appreciated that a size and/or shape of the cabin of each of the plurality of first elevator cars **210** may be determinative of a maximum occupant capacity.

[0049] In the example, the plurality of first elevator cars **210** may include a substantially similar size and/or shape such that the maximum occupant capacities of each are relatively similar. In other examples, the plurality of first elevator cars **210** may include varying sizes and/or shapes, such that the maximum occupant capacity of each may differ relative to one another. In the example, first elevator car **210** may include a total occupancy of one occupant and a maximum occupant capacity of six occupants. Dispatch controller **130** may be configured to determine first elevator car **210** includes an occupancy ratio of approximately 1:6 (e.g., approximately 16.67%).

[0050] Still referring to FIG. 5, at step **316**, dispatch controller **130** may be configured to determine at least one of the plurality of first elevator cars **210** having a maximum occupant capacity that is greater than the remaining plurality of first elevator cars **210**. Dispatch controller **130** may compare the occupancy ratios of each of the plurality of first elevator cars **210** to determine at least one first elevator car **210** having the maximum available occupant capacity. Dispatch controller **130** may assign the call request received at step **302** to the first elevator car **210** of the plurality of first elevator cars **210** (having a travel duration that is less than the threshold duration **148**) that includes the maximum available occupant capacity. At step **318**, dispatch controller may be configured to dispatch the at least one first elevator car **210** to the first location “A.”

[0051] In instances where two or more first elevator cars **210** include a similar occupancy ratio relative to one another, dispatch controller **130** may be configured to compare the minimum duration of each to one another to determine an optimal first elevator car **210** to dispatch. For example, dispatch controller **130** may assign the call request to the first elevator car **210** having the smallest minimum duration to travel to the first location “A.” In other embodiments, dispatch controller **130** may compare the minimum durations of the plurality of first elevator cars **210** to one another even when the occupancy ratios of first elevator cars **210** vary relative to one another. In this instance, dispatch controller **130** may be configured to dispatch at least one first elevator car **210** having the shortest minimum duration despite another one of the plurality of first elevator cars **210** having a greater maximum available occupant capacity.

[0052] It should be appreciated that, in instances where dispatch controller **130** determines only one first elevator car **210** has a minimum duration that is less than the threshold duration **148** (at step **310**), dispatch controller **130** may forgo performance of steps **312** to **316**. In this instance, the first elevator car **210** identified at step **310** may be dispatched to the first location “A” at step **318**. In other embodiments, method **300** may omit steps **312** to **316** entirely such that dispatch controller **130** may be configured to dispatch the at least one first elevator car **210** having the shortest minimum duration, at step **318**.

[0053] Dispatch controller **130** may be configured to communicate with call device **110** to transmit a message to the prospective occupant **20** at the first location “A” (e.g., first floor **204A**). For example, dispatch controller **130** may communicate an identification of the first elevator car **210** dispatched to the first location “A.” In other embodiments,

dispatch controller **130** may identify the first elevator shaft **202** of the plurality of first elevator shafts **202** from which the first elevator car **210** may arrive from. The message may be transmitted via call device **110** in various suitable formats, including, for example, via a display (e.g., a written form, a graphic form, etc.), a speaker (e.g., an audible form), and more.

[0054] Returning to step **310**, in response to determining each of the plurality of first elevator cars **210** has a travel duration that exceeds the threshold duration **148**, dispatch controller **130** may be configured to disregard the plurality of first elevator cars **210** from further consideration. At step **320**, dispatch controller **130** may retrieve motion data **142** from a corresponding motion controller **105** of the plurality of elevator cars **220**, **230** located at various locations in working environment **200** other than the first location “A.” That is, dispatch controller **130** may consider the plurality of second elevator cars **220** (e.g., located adjacent to the second location “B”) and third elevator cars **230** (e.g., located adjacent to the third location “C”) upon determining that none of the plurality of first elevator cars **210** includes a travel duration that is less than the threshold duration **148**.

[0055] Dispatch controller **130** may consider the plurality of elevator cars **220**, **230** despite second elevator cars **220** and third elevator cars **230** being located further from the first location “A” (e.g., a location within working environment **200** proximate to where the call request originated from) than first elevator cars **210**. Dispatch controller **130** may be configured to determine various movement parameters of the plurality of second elevator cars **220** from the motion data **142**, such as, for example, a current location of each second elevator car **220** relative to a respective second elevator shaft **212**, a current travel direction of each second elevator car **220**, a current travel speed of each second elevator car **220**, and more. Dispatch controller **130** may further determine similar movement parameters of the plurality of third elevator cars **230**.

[0056] In response to determining one or more second or third elevator cars **220**, **230** are not traveling toward the first floor **204A**, dispatch controller **130** may be configured to disregard the particular elevator cars **220**, **230** from further consideration. Dispatch controller **130** may be further configured to determine whether a current location of the plurality of second and third elevator cars **220**, **230** are located prior to first floor **204A**. In the example, working environment **200** includes second elevator car **220** stationary at fourth floor **204D**, and third elevator car **230** stationary at second floor **204B**, such that dispatch controller **130** may determine that second elevator car **220** and third elevator car **230** are located before first floor **204A**.

[0057] Still referring to FIG. 5, at step **322**, dispatch controller **130** may be configured to determine how many, if any, intermediate stops second elevator cars **220** are expected to have between a respective current location and the second location “B” (e.g., on first floor **204A**). Dispatch controller **130** may similarly determine the number of calls assigned to third elevator cars **230**, and that have a (pick up) location positioned between a respective current location of each third elevator car **230** and the third location “C” (e.g., on first floor **204A**). In the example, second elevator car **220** may include an assigned call at second floor **204B**, and third elevator car **230** may not include any assigned calls. Accordingly, dispatch controller **130** may determine that second

elevator car **220** includes one stop between the current location of second elevator car **220** and the second location “B.”

[0058] At step **324**, dispatch controller **130** may be configured to analyze the motion data **142** (step **322**) and the call assignment data **146** (step **324**) of the plurality of elevator cars **220**, **230** to determine a minimum duration for each elevator car **220**, **230** to travel from a respective current location to the second and/or third location “B, C.” For example, dispatch controller **130** may analyze a travel distance between a respective current location of each elevator car **220**, **230** and the corresponding second and/or third location “B, C” when determining the minimum duration. Dispatch controller **130** may further analyze a travel speed of each elevator car **220**, **230** when determining the minimum duration. By further example, dispatch controller **130** may determine an additional duration required to travel to the respective location “B, C,” when elevator car **220**, **230** includes a preexisting call assignment. Dispatch controller **130** may assess the number of stops assigned to each elevator car **220**, **230** when determining the minimum duration for each elevator car **220**, **230**.

[0059] Still referring to FIG. 5, at step **326**, dispatch controller **130** may be further configured to modify the minimum duration computation for elevator cars **220**, **230** based on a distance the prospective occupant **20** may be required to travel (e.g., walk) from the first location “A” to each of the second location “B” and third location “C.” Dispatch controller **130** may include data (e.g., duration data **150**) indicative of the time required to travel from the first location “A,” at which the call request originated from, to the second location “B” at which second elevator cars **220** are positioned, based on the distance between the first location “A” and the second location “B.” Dispatch controller **130** may be configured to consider the distance required for the prospective occupant **20** to travel between locations “A, B” when determining the minimum duration (e.g., a second duration) for each of the plurality of second elevator cars **220**.

[0060] Dispatch controller **130** may further include data indicative of the time required for prospective passengers **20** to travel between the first location “A” and the third location “C,” at which third elevator cars **230** are positioned, based on the distance between the first location “A” and the third location “C.” Dispatch controller **130** may be configured to consider the distance required for the prospective occupant **20** to travel between locations “A, C” when determining the minimum duration (e.g., a third duration) for each of the plurality of third elevator cars **230**.

[0061] Accordingly, dispatch controller **130** may be configured to modify (e.g., increase) the minimum duration of the plurality of elevator cars **220**, **230** based on the distance between first location “A” and the respective second location “B” or third location “C” when determining the minimum duration for elevator cars **220**, **230**. In other embodiments, dispatch controller **130** may be configured to compute the distance between the first location “A” and at least one of the second location “B” or the third location “C.”

[0062] At step **328**, dispatch controller **130** may be configured to compare the minimum duration of the plurality of elevator cars **220**, **230** to the threshold duration **148**. In response to determining two or more of the plurality of elevator cars **220**, **230** have a minimum duration that is less

than the threshold duration **148** at step **328**, dispatch controller **130** may be configured to determine the number of occupants **10** within each of the two or more elevator cars **220**, **230** at step **330**. For example, dispatch controller **130** may be configured to determine the number of occupants **10** within each elevator car **220**, **230** (having a minimum duration less than the threshold duration **148**) by retrieving occupant data **144** from the respective counter device **125** of each elevator car **220**, **230**.

[0063] Still referring to FIG. 5, at step **332**, dispatch controller **130** may be configured to determine an occupancy ratio of each of the plurality of elevator cars **220**, **230** based on at least the number of occupants **10** detected within the elevator car (step **330**) and a maximum occupant capacity of each elevator car. In the example, second elevator car **220** may include a total occupancy of two occupants and a maximum occupant capacity of six occupants, and third elevator car **230** may include a total occupancy of one occupant and a maximum occupant capacity of five occupants. Dispatch controller **130** may be configured to determine second elevator car **220** includes an occupancy ratio of 2:6 (e.g., approximately 33.33%), and third elevator car **230** includes an occupancy ratio of 1:5 (e.g., approximately 20%).

[0064] At step **334**, dispatch controller **130** may be configured to determine at least one of the plurality of elevator cars **220**, **230** having a maximum occupant capacity that is greater than the remaining elevator cars **220**, **230**. Dispatch controller **130** may compare the occupancy ratios of each of the plurality of elevator cars **220**, **230** to determine at least one second elevator car **220** or third elevator car **230** having the maximum available occupant capacity. Dispatch controller **130** may assign the call request received at step **302** to the elevator car **220**, **230** (having a travel duration that is less than the threshold duration **148**) that includes the maximum available occupant capacity. At step **336**, dispatch controller **130** may dispatch the elevator car **220**, **230** having the maximum available occupant capacity to the corresponding location “B, C” on first floor **204A**.

[0065] In instances where two or more second elevator cars **220** and/or third elevator cars **230** include a similar occupancy ratio relative to one another, dispatch controller **130** may compare the minimum duration of each to determine an optimal elevator car **220**, **230** to dispatch. For example, dispatch controller **130** may dispatch the second elevator car **220** having the smallest minimum duration for traveling to the second location “B,” or the third elevator car **230** having the smallest minimum duration for traveling to the third location “C.” In other embodiments, dispatch controller **130** may compare the minimum durations of the plurality of elevator cars **220**, **230** to one another even when the occupancy ratios of elevator cars **220**, **230** vary relative to one another. In this instance, dispatch controller **130** may be configured to dispatch at least one elevator car **220**, **230** having the shortest minimum duration despite another one of the plurality of elevator cars **220**, **230** having a greater maximum available occupant capacity.

[0066] It should be appreciated that, in instances where dispatch controller **130** determines only one second elevator car **220** or third elevator car **230** has a minimum duration that is less than the threshold duration **148** (at step **328**), dispatch controller **130** may forgo performance of steps **330** to **334**. In this instance, the elevator car **220**, **230** identified at step **328** may be dispatched at step **336**. In other embodiments,

ments, method 300 may omit steps 330 to 334 entirely such that dispatch controller 130 may be configured to dispatch the at least second elevator car 220 or third elevator car 230 having the shortest minimum duration, at step 336.

[0067] Returning to step 328, in response to determining each of the plurality of second elevator cars 220 and third elevator cars 230 have a travel duration that exceeds the threshold duration 148, dispatch controller 130 may be configured to disregard the plurality of second elevator cars 220 and third elevator cars 230 from further consideration. In this instance, dispatch controller 130 may assign the call request received at step 302 to one of the plurality of first elevator cars 210, despite each of the first elevator cars 210 having a minimum duration that exceeds the threshold duration 148. In this instance, dispatch controller 130 may determine the number of occupants present (step 312), the occupancy ratio (step 314), and the maximum available occupancy ratio (step 316) of each of the plurality of first elevator cars 210. At step 318, dispatch controller 130 may dispatch the first elevator car 210 of the plurality of first elevator cars 210 having the maximum available occupancy. Alternatively, dispatch controller 130 may omit steps 312 to 316, and dispatch the at least one first elevator car 210 having the shortest minimum duration, at step 318.

[0068] All technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which this disclosure belongs unless clearly indicated otherwise. As used herein, the singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.

[0069] The above description is illustrative and is not intended to be restrictive. One of ordinary skill in the art may make numerous modifications and/or changes without departing from the general scope of the disclosure. For example, and as has been described, the above-described embodiments (and/or aspects thereof) may be used in combination with each other. Additionally, portions of the above-described embodiments may be removed without departing from the scope of the disclosure. In addition, modifications may be made to adapt a particular situation or material to the teachings of the various embodiments without departing from their scope. Many other embodiments will also be apparent to those of skill in the art upon reviewing the above description.

What is claimed is:

1. A method for dispatching an elevator car, the method comprising:

determining a first duration for each of a plurality of first elevator cars to travel from a current location to a first location;

dispatching at least one of the plurality of first elevator cars to the first location when the first duration of the at least one of the plurality of first elevator cars is less than a threshold duration;

determining a second duration that includes a total duration for (i) an occupant to travel from the first location to a second location, and (ii) each of a plurality of second elevator cars to travel from the current location to the second location, when the first duration for each of the plurality of first elevator cars exceeds the threshold duration; and

dispatching at least one of the plurality of first elevator cars to the first location when the second duration for each of the plurality of second elevator cars exceeds the threshold duration.

2. The method of claim 1, further comprising:

dispatching the at least one of the plurality of first elevator cars to the first location when the first duration of the at least one first elevator car is less than the first duration of a remaining plurality of first elevator cars; and

dispatching the at least one of the plurality of second elevator cars to the second location when the second duration of the at least one second elevator car is less than the second duration of a remaining plurality of second elevator cars.

3. The method of claim 1, wherein determining the first duration for each of the plurality of first elevator cars comprises:

receiving motion data from each of the plurality of first elevator cars; and

determining a travel direction, a travel speed, a number of preassigned calls, or the current location of each of the plurality of first elevator cars from the motion data.

4. The method of claim 3, wherein determining the number of preassigned calls includes calls from locations positioned between the current location of each of the plurality of first elevator cars and the first location.

5. The method of claim 3, further comprising:

determining a distance between the first location and the current location of each of the plurality of first elevator cars; and

dispatching the at least one of the plurality of first elevator cars having a shortest distance to the first location.

6. The method of claim 1, wherein prior to dispatching the at least one of the plurality of first elevator cars to the first location, the method comprises:

comparing the first duration of each of the plurality of first elevator cars to the threshold duration; and

determining an occupant capacity for each of the plurality of first elevator cars with the first duration that is less than the threshold duration.

7. The method of claim 6, wherein determining the occupant capacity for each of the plurality of first elevator cars comprises:

determining a number of occupants located within each of the plurality of first elevator cars.

8. The method of claim 7, further comprising:

determining a ratio between the number of occupants located within each of the plurality of first elevator cars and a maximum occupancy capacity of each of the plurality of first elevator cars.

9. The method of claim 8, further comprising:

comparing the ratio of each of the plurality of first elevator cars to one another; and

determining the at least one of the plurality of first elevator cars has an available ratio that is greater than the ratio of a remaining plurality of first elevator cars.

10. The method of claim 9, wherein dispatching the at least one of the plurality of first elevator cars to the first location comprises:

dispatching the at least one of the plurality of first elevator cars having a greatest available ratio relative to the remaining plurality of first elevator cars.

11. The method of claim 1, wherein when the second duration for at least one of the plurality of second elevator cars is less than the threshold duration, the method comprises:

dispatching the at least one of the plurality of second elevator cars to the second location such that at least one of the plurality of first elevator cars is not dispatched to the first location.

12. The method of claim 1, wherein determining the second duration for each of the plurality of second elevator cars further comprises:

receiving motion data from each of the plurality of second elevator cars; and

determining a travel direction, a travel speed, a number of preassigned calls, or the current location of each of the plurality of second elevator cars from the motion data.

13. The method of claim 1, wherein prior to dispatching the at least one of the plurality of first elevator cars to the first location, the method comprises:

comparing the first duration of each of the plurality of first elevator cars to one another; and

dispatching the at least one of the plurality of first elevator cars to the first location when the first duration of the at least one first elevator car is less than the first duration of a remaining plurality of first elevator cars.

14. The method of claim 1, wherein prior to dispatching the at least one of the plurality of first elevator cars to the first location, the method comprises:

determining a number of occupants located within each of the plurality of first elevator cars; and

dispatching the at least one of the plurality of first elevator cars having the least number of occupants relative to a remaining plurality of first elevator cars.

15. The method of claim 1, wherein prior to determining the first duration for each of the plurality of first elevator cars, the method comprises:

receiving a call for the elevator car, wherein the call is from the first location of a plurality of locations.

16. A system for dispatching an elevator car, comprising: at least one motion controller operably coupled to a plurality of elevator cars, the at least one motion controller is configured to determine a current location of the plurality of elevator cars, wherein the plurality of elevator cars includes a first subset and a second subset; and

a dispatch controller operably coupled to the at least one motion controller of the plurality of elevator cars, such that the dispatch controller receives data indicative of the current location of the plurality of elevator cars, wherein the dispatch controller is configured to:

determine a first duration for each of the first subset of elevator cars to travel from the current location to a first location;

dispatch, in response to the first duration of at least one of the first subset of elevator cars not exceeding a threshold duration, the at least one elevator car of the first subset to the first location;

determine, in response to the first duration for each of the first subset of elevator cars exceeding the threshold duration, a second duration that includes a total duration for (i) an occupant to travel from the first location to a second location, and (ii) each of the

second subset of elevator cars to travel from the current location to the second location; and

dispatch at least one of the plurality of first elevator cars to the first location when the second duration for each of the plurality of second elevator cars exceeds the threshold duration.

17. The system of claim 16, further comprising at least one counter device operably coupled to the plurality of elevator cars, the at least one counter device is configured to determine a number of occupants within each of the plurality of elevator cars;

wherein the dispatch controller is operably coupled to the at least one counter device of the plurality of elevator cars, such that the dispatch controller receives data indicative of the number of occupants within the plurality of elevator cars.

18. The system of claim 17, wherein the dispatch controller is configured to:

determine a total occupancy for each of the plurality of elevator cars; and

dispatch, in response to the total occupancy of at least one of the first subset of elevator cars being less than the total occupancy of a remainder of the first subset of elevator cars, the at least one elevator car of the first subset to the first location.

19. The system of claim 16, wherein the dispatch controller is configured to:

compare the first duration of each of the first subset of elevator cars to one another when the second duration for each of the second subset of elevator cars exceeds the threshold duration; and

dispatch the at least one of the first subset of elevator cars to the first location when the first duration of the at least one first subset of elevator cars is less than the first duration of a remainder of the first subset of elevator cars.

20. A system for controlling traffic flow of a plurality of elevator cars, comprising:

a processor; and

a memory storing instructions that, when executed by the processor, causes the processor to perform operations including:

determining a first duration for each of a plurality of first elevator cars to travel from a current location to a first location;

dispatching at least one of the plurality of first elevator cars to the first location when the first duration of the at least one first elevator car is less than a threshold duration;

determining a second duration that includes a total duration for (i) an occupant to travel from the first location to a second location, and (ii) each of a plurality of second elevator cars to travel from the current location to the second location, when the first duration for each of the plurality of first elevator cars exceeds the threshold duration; and

dispatching at least one of the plurality of first elevator cars to the first location when the second duration for each of the plurality of second elevator cars exceeds the threshold duration.

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